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|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------|
|  <b>Marwadi University</b> | <b>Marwadi University</b><br><b>Faculty of Technology</b><br><b>Department of Information and Communication Technology</b> |                                  |
| <b>Subject: Artificial Intelligence (01CT0616)</b>                                                          | <b>Aim: To understand the process of convolution over the image and apply over the classification problem</b>              |                                  |
| <b>Experiment No: 4</b>                                                                                     | <b>Date:</b>                                                                                                               | <b>Enrolment No: 92301733043</b> |

**Aim:** To understand the process of convolution over the image and apply over the classification problem

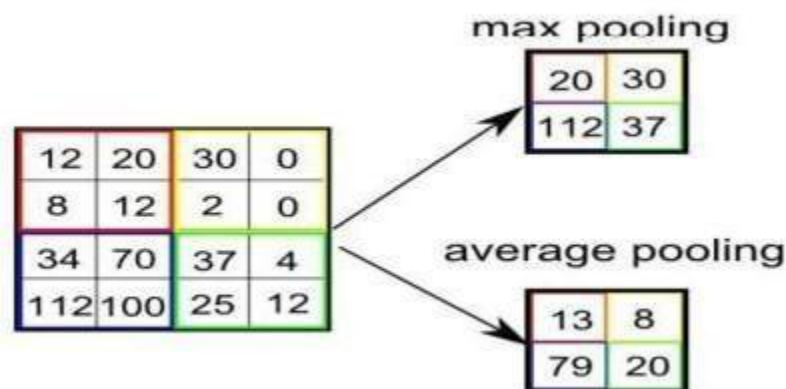
**IDE:** Google Colab

### **Theory:**

Convolutional Neural Networks (CNN) are complex feed forward neural networks. CNNs are used for image classification and recognition because of its high accuracy. There are three types of layers in a convolutional neural network: i. Convolutional layer ii. Pooling layer iii. Fully connected layer. Each of these layers has different parameters that can be optimized and performs a different task on the input data.

### **What is Pooling Layer?**

Pooling layer is responsible for reducing the spatial size of the Convolved Feature. This is to decrease the computational power required to process the data through dimensionality reduction. There are two types of Pooling: i. Average Pooling. ii. Max Pooling. Max Pooling returns the maximum value from the portion of the image covered by the Kernel. On the other hand, Average Pooling returns the average of all the values from the portion of the image covered by the Kernel. Max Pooling also performs as a Noise Suppressant. It discards the noisy activations altogether and also performs de-noising along with dimensionality reduction. On the other hand, Average Pooling simply performs dimensionality reduction as a noise suppressing mechanism. Hence, we can say that Max Pooling performs a lot better than Average Pooling.



### **What is Convolutional Layer?**

Convolutional layers are the major building blocks used in convolutional neural networks. A convolution is the simple application of a filter to an input that results in an activation. Repeated application of the same filter to an input results in a map of activations called a feature map, indicating the locations and strength of a detected feature in an input, such as an image. A convolutional layer contains a set of filters whose parameters need to be learned. The height and weight of the filters are smaller than those of the input volume. Each filter is convolved with the input volume to compute an activation map made of neurons.

|                                                                                                             |                                                                                                                            |                                  |
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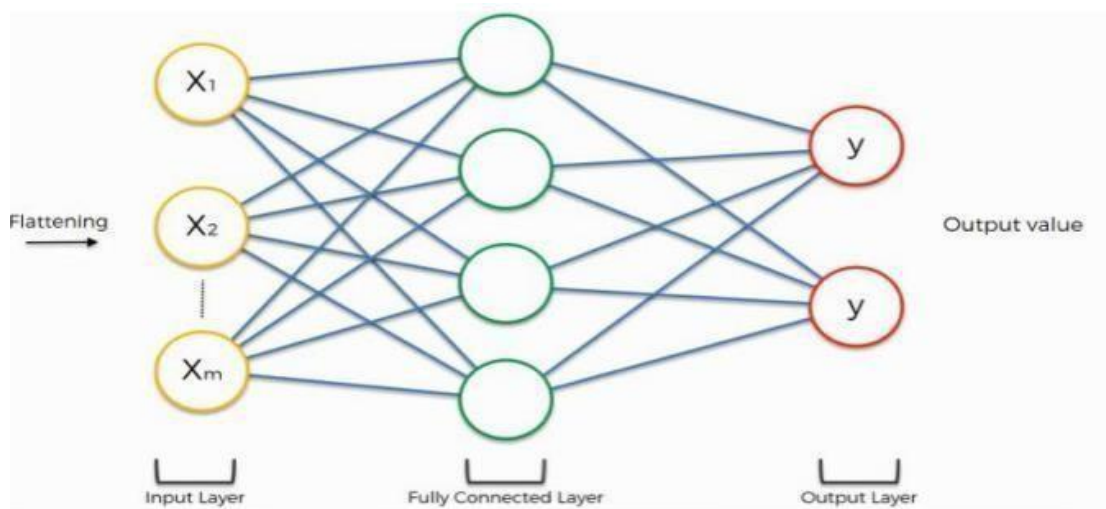
### What is Fully Connected Layer?

A fully connected layer that takes the output of convolution/pooling and predicts the best label to describe the image. We have three layers in the full connection step: i. Input layer, ii. Fully-connected layer, iii. Output layer.

**Input Layer:** It takes the output of the previous layers, “flattens” them and turns them into a single vector that can be an input for the next stage.

**Fully Connected Layer:** It takes the inputs from the feature analysis and applies weights to predict the correct label.

**Output Layer:** It gives the final probabilities for each label.



**ReLU Layer:** ReLU is an activation function. Rectified Linear Unit (ReLU) transform function only activates a node if the input is above a certain quantity, while the input is below zero, the output is zero, but when the input rises above a certain threshold, it has a linear relationship with the dependent variable. The main aim is to remove all the negative values from the convolution. All the positive values remain the same but all the negative values get changed to zero.

### Methodology:

1. Load the basic libraries and packages
2. Load the dataset
3. Analyse the dataset
4. Normalize the data
5. Pre-process the data
6. Visualize the Data

|                                                                                                                |                                                                                                                            |                                  |
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7. Write the CNN model function
8. Write the Cost Function
9. Write the Gradient Descent optimization algorithm
10. Apply the training over the dataset to minimize the loss
11. Observe the cost function vs iterations learning curve

### **Program (Code):**

```
#import libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import keras
from keras.datasets import fashion_mnist
import keras.models as models
import keras.layers as layers
from keras import regularizers
from keras.layers import Dropout
import tensorflow as tf
from tensorflow.keras.models import Sequential, Model
from tensorflow.keras.layers import Dense, Activation, Flatten, Dropout, Conv2D, MaxPooling2D, LeakyReLU
from tensorflow.keras.utils import to_categorical #for the categorical data

(train_X, train_Y), (test_X, test_Y) = fashion_mnist.load_data()

plt.imshow(train_X[0])
print(train_Y[0])

print(train_X.shape, train_Y.shape)
print(test_X.shape, test_Y.shape)

classes = np.unique(train_Y)
nClasses = len(classes)
print(classes)
print(nClasses)

train_X = train_X.reshape(-1, 28, 28, 1)
test_X = test_X.reshape(-1, 28, 28, 1)
```

|                                                                                                                |                                                                                                                            |                                  |
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```
print(train_X.shape,train_Y.shape)
print(test_X.shape, test_Y.shape)
```

```
train_X=train_X.astype('float32')
test_X=test_X.astype('float32')
train_X=train_X/255
test_X=test_X/255
```

```
train_Y_one_hot=to_categorical(train_Y)
test_Y_one_hot=to_categorical(test_Y)
```

```
train_Y[0]
train_Y_one_hot[0]
```

```
from sklearn.model_selection import train_test_split
train_X,valid_X,train_label,valid_label=train_test_split(train_X,train_Y_one_hot,test_size=0.2,random_state=13)
```

```
train_X.shape,valid_X.shape,train_label.shape,valid_label.shape
```

```
batch_size=64
epochs=20
num_classes=10
```

```
model=tf.keras.Sequential()
```

```
model.add(tf.keras.layers.Conv2D(32,(3,3),activation="linear",padding='same'))
model.add(tf.keras.layers.LeakyReLU(alpha=0.1))
model.add(tf.keras.layers.MaxPooling2D(pool_size=(2,2),padding='same'))
```

```
model.add(tf.keras.layers.Conv2D(64,(3,3),activation="linear",padding='same'))
model.add(tf.keras.layers.LeakyReLU(alpha=0.1))
model.add(tf.keras.layers.MaxPooling2D(pool_size=(2,2),padding='same'))
```

```
model.add(tf.keras.layers.Conv2D(128,(3,3),activation="linear",padding='same'))
model.add(tf.keras.layers.LeakyReLU(alpha=0.1))
model.add(tf.keras.layers.MaxPooling2D(pool_size=(2,2),padding='same'))
```

```
model.add(tf.keras.layers.Flatten())
```

```
model.add(tf.keras.layers.Dense(128,activation='linear'))
model.add(tf.keras.layers.LeakyReLU(alpha=0.1))
model.add(tf.keras.layers.Dense(num_classes,activation='softmax'))
```

|                                                                                                                |                                                                                                                            |                                  |
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```

model.compile(loss=tf.keras.losses.categorical_crossentropy,optimizer=tf.keras.optimizers.Adam(),metrics=['accuracy'])

model_train=model.fit(train_X,train_label,batch_size=64,epochs=20,verbose=1,validation_data=(valid_X,valid_label))

testing_evaluation=model.evaluate(test_X,test_Y_one_hot)
testing_evaluation

model_train.history

accuracy=model_train.history['accuracy']
val_accuracy=model_train.history['val_accuracy']
loss=model_train.history['loss']
val_loss=model_train.history['val_loss']
epochs=range(len(accuracy))

plt.plot(epochs,accuracy,label='training accuracy')
plt.plot(epochs,val_accuracy,label='validation accuracy')
plt.title('evaluating based on accuracy')
plt.legend()
plt.show()

plt.plot(epochs,loss,label='training loss')
plt.plot(epochs,val_loss,label='validation loss')
plt.title('evaluating based on loss')
plt.legend()
plt.show()

regmodel=tf.keras.Sequential()

regmodel.add(tf.keras.layers.Conv2D(32,(3,3),activation="linear",padding='same'))
regmodel.add(tf.keras.layers.LeakyReLU(alpha=0.1))
regmodel.add(tf.keras.layers.MaxPooling2D(pool_size=(2,2),padding='same'))
regmodel.add(tf.keras.layers.Dropout(0.25))

regmodel.add(tf.keras.layers.Conv2D(64,(3,3),activation="linear",padding='same'))
regmodel.add(tf.keras.layers.LeakyReLU(alpha=0.1))
regmodel.add(tf.keras.layers.MaxPooling2D(pool_size=(2,2),padding='same'))
regmodel.add(tf.keras.layers.Dropout(0.25))

regmodel.add(tf.keras.layers.Conv2D(128,(3,3),activation="linear",padding='same'))

```

|                                                                                                             |                                                                                                                            |                                  |
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```

regmodel.add(tf.keras.layers.LeakyReLU(alpha=0.1))
regmodel.add(tf.keras.layers.MaxPooling2D(pool_size=(2,2),padding='same'))
regmodel.add(tf.keras.layers.Dropout(0.40))

regmodel.add(tf.keras.layers.Flatten())
regmodel.add(tf.keras.layers.Dense(128,activation='linear'))
regmodel.add(tf.keras.layers.LeakyReLU(alpha=0.1))
regmodel.add(tf.keras.layers.Dropout(0.3))
regmodel.add(tf.keras.layers.Dense(num_classes,activation='softmax'))

regmodel.compile(loss=tf.keras.losses.categorical_crossentropy,optimizer=tf.keras.optimizers.Adam(),metrics=['accuracy'])

reg_model_train=regmodel.fit(train_X,train_label,batch_size=64,epochs=20,verbose=1,validation_data=(valid_X,valid_label))

accuracy = reg_model_train.history['accuracy']
val_accuracy = reg_model_train.history['val_accuracy']
epochs = range(len(accuracy))

plt.figure(figsize=(8,5))
plt.plot(epochs, accuracy, label='Training Accuracy')
plt.plot(epochs, val_accuracy, label='Validation Accuracy')

plt.title('Training and Validation Accuracy w.r.t Epochs (After Regularization)')
plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.legend()

plt.show()

loss = reg_model_train.history['loss']
val_loss = reg_model_train.history['val_loss']
epochs = range(len(loss))

plt.figure(figsize=(8,5))
plt.plot(epochs, loss, label='Training Loss')
plt.plot(epochs, val_loss, label='Validation Loss')

plt.title('Training and Validation Loss w.r.t Epochs (After Regularization)')
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.legend()

```

|                                                                                                             |                                                                                                                            |                                  |
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```
plt.show()
```

```
predictions = regmodel.predict(test_X)
predicted_classes = np.argmax(predictions, axis=1)
```

```
true_classes = test_Y
```

```
correct = np.where(predicted_classes == true_classes)[0]
incorrect = np.where(predicted_classes != true_classes)[0]
```

```
print("Correct predictions:", len(correct))
print("Incorrect predictions:", len(incorrect))
```

```
plt.figure(figsize=(10,10))
```

```
for i, correct_index in enumerate(correct[:9]):
    plt.subplot(3,3,i+1)
    plt.imshow(test_X[correct_index].reshape(28,28), cmap='gray')
    plt.title("Original: {}, Predicted: {}".format(
        true_classes[correct_index],
        predicted_classes[correct_index]
    ))
    plt.axis('off')
```

```
plt.tight_layout()
plt.show()
```

```
plt.figure(figsize=(10,10))
```

```
for i, correct_index in enumerate(correct[:9]):
    plt.subplot(3,3,i+1)
    plt.imshow(test_X[correct_index].reshape(28,28), cmap='gray')
    plt.title("Original: {}, Predicted: {}".format(
        true_classes[correct_index],
        predicted_classes[correct_index]
    ))
    plt.axis('off')
```

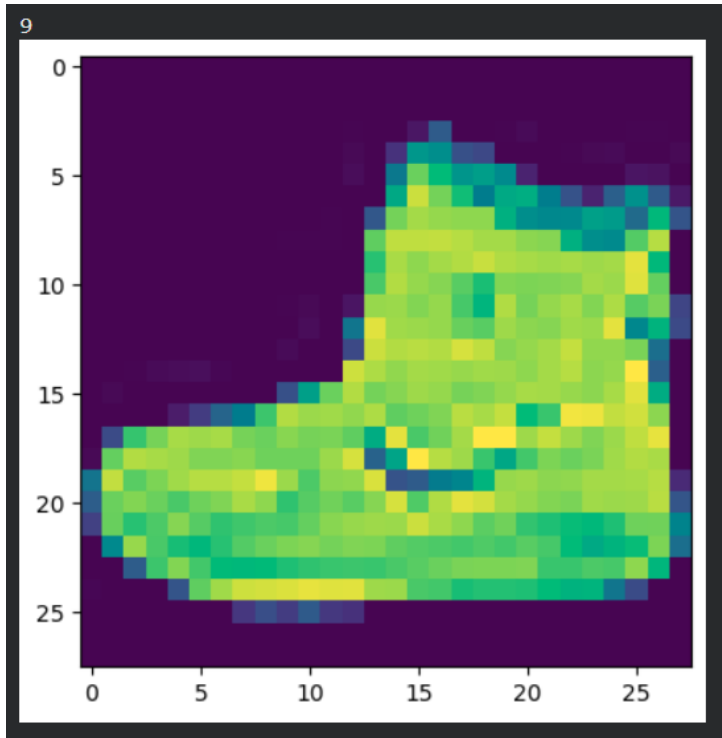
```
plt.tight_layout()
plt.show()
```

|                                                                                                             |                                                                                                                            |                                  |
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## Results:

To be attached with

- Training dataset

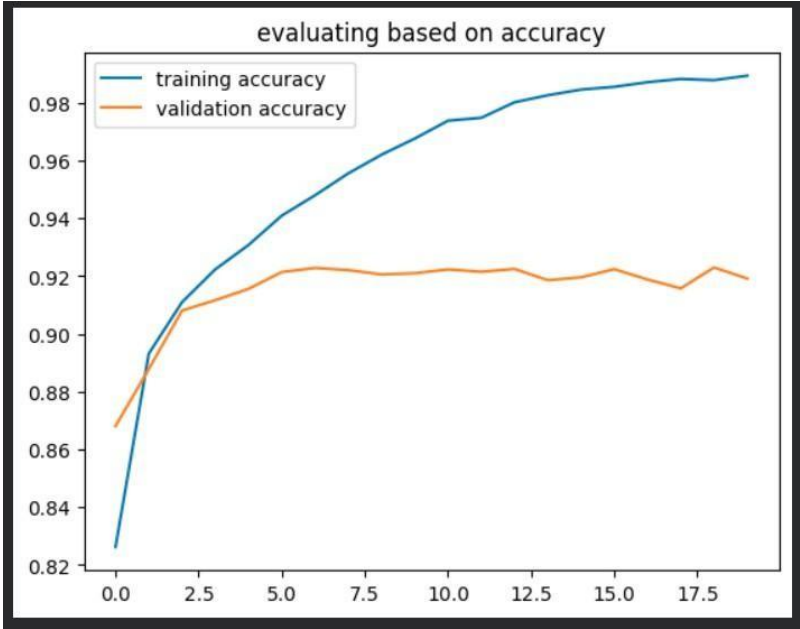


- Model summary

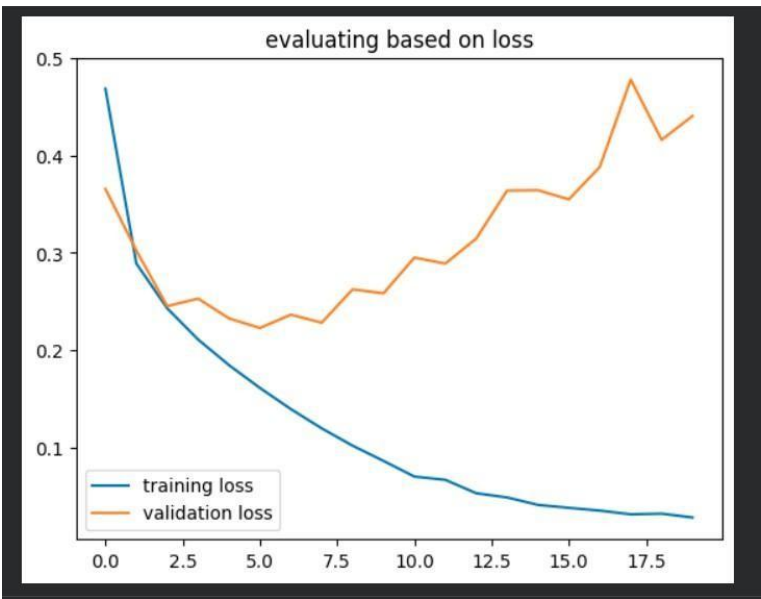
| Model: "sequential"                 |                  |         |
|-------------------------------------|------------------|---------|
| Layer (type)                        | Output Shape     | Param # |
| conv2d (Conv2D)                     | (64, 28, 28, 32) | 320     |
| leaky_re_lu (LeakyReLU)             | (64, 28, 28, 32) | 0       |
| max_pooling2d (MaxPooling2D)        | (64, 14, 14, 32) | 0       |
| conv2d_1 (Conv2D)                   | (64, 14, 14, 64) | 18,496  |
| leaky_re_lu_1 (LeakyReLU)           | (64, 14, 14, 64) | 0       |
| max_pooling2d_1 (MaxPooling2D)      | (64, 7, 7, 64)   | 0       |
| conv2d_2 (Conv2D)                   | (64, 7, 7, 128)  | 73,856  |
| leaky_re_lu_2 (LeakyReLU)           | (64, 7, 7, 128)  | 0       |
| max_pooling2d_2 (MaxPooling2D)      | (64, 4, 4, 128)  | 0       |
| flatten (Flatten)                   | (64, 2048)       | 0       |
| dense (Dense)                       | (64, 128)        | 262,272 |
| leaky_re_lu_3 (LeakyReLU)           | (64, 128)        | 0       |
| dense_1 (Dense)                     | (64, 10)         | 1,290   |
| Total params: 1,068,704 (4.08 MB)   |                  |         |
| Trainable params: 356,234 (1.36 MB) |                  |         |
| Non-trainable params: 0 (0.00 B)    |                  |         |
| Optimizer params: 712,470 (2.72 MB) |                  |         |

|                                                                                                             |                                                                                                                            |                                  |
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c. Training and validation accuracy w.r.t epochs before regularization

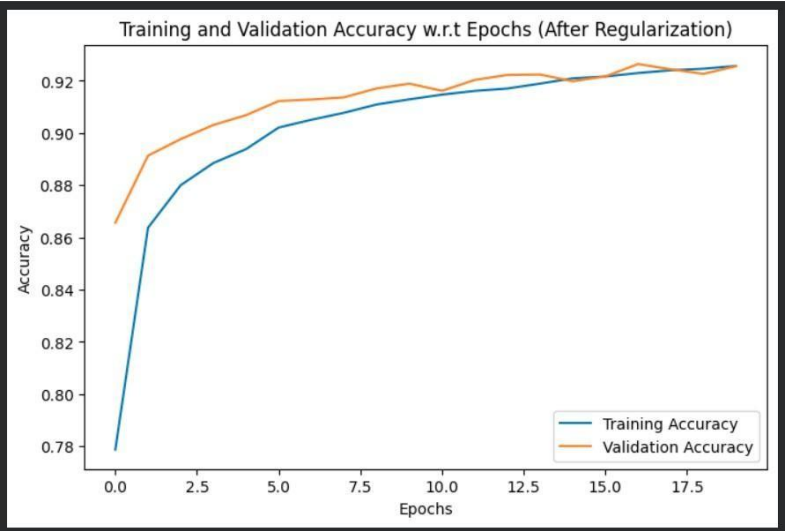


d. Training and validation loss w.r.t epochs before regularization

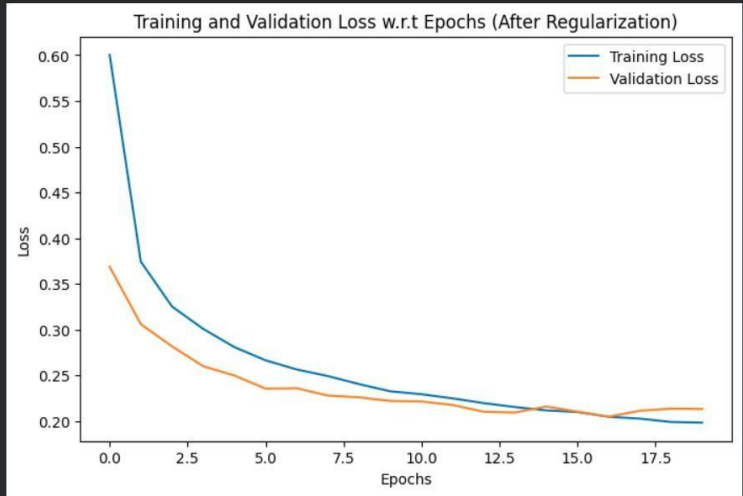


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e. Training and validation accuracy w.r.t epochs after regularization



f. Training and validation loss w.r.t epochs after regularization



|                                                                                                             |                                                                                                                            |                                  |
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g. Original v/s predicted labels for correct predicted observations



h. Original v/s predicted labels for incorrect predicted observations



|                                                                                                                |                                                                                                                            |                                  |
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### **Pre Lab Exercise:**

- a. What is Convolution process? Explain giving example.

Convolution is a process used in Convolutional Neural Networks (CNNs) to extract important features from an image. It works by sliding a small matrix called a filter (kernel) over the input image and performing element-wise multiplication and summation to produce a feature map. This helps detect patterns such as edges, textures, and shapes.

Suppose we have a **3×3 filter** applied to a part of an image matrix.

Input (3×3 region):

```
1 1 1
0 1 1
0 0 1
```

Filter:

```
1 0 1
0 1 0
1 0 1
```

Multiply corresponding elements and add them:

```
(1×1) + (1×0) + (1×1) +
(0×0) + (1×1) + (1×0) +
(0×1) + (0×0) + (1×1) = 4
```

The result **4** becomes the first value in the **feature map**.

- b. What are different layers in CNN model?

- Convolutional Layer  
**This layer applies filters (kernels) to the input image to extract important features such as edges, textures, and patterns. It produces a feature map.**
- Activation Layer (ReLU / LeakyReLU)  
**This layer introduces non-linearity into the model so that the network can learn complex patterns.**
- Pooling Layer (Max Pooling / Average Pooling)  
**This layer reduces the spatial size of the feature map, which decreases computation and helps prevent overfitting.**

|                                                                                                                |                                                                                                                            |                                  |
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- Flatten Layer  
**This layer converts the 2D feature maps into a 1D vector so it can be fed into the fully connected layers.**
- Fully Connected (Dense) Layer  
**This layer performs the final classification by combining the features learned by previous layers.**
- Dropout Layer (Regularization)  
This layer randomly disables some neurons during training to **reduce overfitting** and improve generalization.

c. What is the requirement of the pooling layer in the CNN model?

- The pooling layer is used to reduce the spatial size (width and height) of feature maps **generated** by the convolution layer. This helps decrease the number of parameters and computational cost in the neural network.
- Pooling also helps in removing unnecessary details while preserving important features, which improves the model's ability to generalize and reduces overfitting.

d. What is the requirement of the use of ReLU activation function after convolution step?

- The ReLU (Rectified Linear Unit) activation function is applied after the convolution layer to introduce non-linearity into the CNN model. Without activation functions, the neural network would behave like a simple linear model and would not be able to learn complex patterns.
- ReLU works by converting negative values to zero while keeping positive values unchanged.

|                                                                                                             |                                                                                                                            |                                  |
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### **Observation and Result Analysis:**

#### a. Nature of the dataset

The dataset used in this model is the Fashion-MNIST dataset, which is a collection of grayscale images of clothing items used for image classification tasks.

Key characteristics:

- The dataset contains 70,000 images in total.
- 60,000 images are used for training and 10,000 images for testing.
- Each image has a size of  $28 \times 28$  pixels.
- Images are grayscale, meaning pixel values range from 0 to 255.
- There are 10 different classes representing clothing items such as T-shirt, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag, and Ankle boot.

#### b. Training Process without regularization

A Convolutional Neural Network (CNN) model is trained on the Fashion-MNIST dataset without using any regularization techniques such as Dropout.

Process:

1. The dataset is loaded and pre-processed by reshaping images to  $28 \times 28 \times 1$  and normalizing pixel values between 0 and 1.
2. The labels are converted into one-hot encoded vectors for multi-class classification.
3. The training dataset is further split into training and validation sets.
4. A CNN model is built using Convolution layers, LeakyReLU activation, MaxPooling layers, Flatten layer, and Dense layers.
5. The model is compiled using the Adam optimizer and categorical cross-entropy loss function.
6. The model is trained for 20 epochs with a batch size of 64.
7. During training, training accuracy, validation accuracy, training loss, and validation loss are recorded to evaluate the model performance.

#### c. After regularization in the training Process

After applying regularization, the CNN model is trained again to improve generalization and reduce overfitting. In this model, Dropout layers are added after convolution and dense layers.

Process:

|                                                                                                                |                                                                                                                            |                                  |
|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------|
|  <b>Marwadi</b><br>University | <b>Marwadi University</b><br><b>Faculty of Technology</b><br><b>Department of Information and Communication Technology</b> |                                  |
| <b>Subject: Artificial Intelligence (01CT0616)</b>                                                             | <b>Aim: To understand the process of convolution over the image and apply over the classification problem</b>              |                                  |
| <b>Experiment No: 4</b>                                                                                        | <b>Date:</b>                                                                                                               | <b>Enrolment No: 92301733043</b> |

1. The same preprocessed Fashion-MNIST dataset is used for training and validation.
2. A CNN model is created with Convolution, LeakyReLU activation, and MaxPooling layers.
3. Dropout layers (e.g., 0.25, 0.40, 0.30) are added after some layers to randomly deactivate neurons during training.
4. The model is compiled using the Adam optimizer and categorical cross-entropy loss function.
5. The model is trained for 20 epochs with a batch size of 64.

d. Observation over the Learning Curves

- The learning curves show the training and validation accuracy and loss across epochs.
- Before regularization, the training accuracy increases continuously, while the validation accuracy increases slowly and may start to stabilize or decrease after some epochs. This indicates that the model begins to overfit the training data. Similarly, the training loss decreases steadily, but the validation loss may remain higher or fluctuate.
- After applying regularization (Dropout), the gap between training and validation accuracy becomes smaller, and the validation loss becomes more stable. This shows that the model is able to generalize better on unseen data.

|                                                                                                                |                                                                                                                            |                                  |
|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------|
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| <b>Experiment No: 4</b>                                                                                        | <b>Date:</b>                                                                                                               | <b>Enrolment No: 92301733043</b> |

### **Post Lab Exercise:**

- a. Why CNN is preferred over ANN for images

**Convolutional Neural Networks (CNNs)** are preferred over **Artificial Neural Networks (ANNs)** for image processing because CNNs are specifically designed to handle **spatial data like images**.

#### **Reasons:**

1. **Automatic Feature Extraction**

CNNs automatically detect important features such as **edges, shapes, and textures**, while ANN requires manual feature extraction.

2. **Preserves Spatial Relationships**

CNNs maintain the **spatial structure of images**, whereas ANN converts images into a 1D vector, losing spatial information.

3. **Fewer Parameters**

CNN uses **shared weights and filters**, which significantly reduces the number of parameters compared to ANN.

4. **Better Performance for Images**

CNN models are more effective for **image classification, object detection, and pattern recognition** tasks.

- b. Can CNN be applied over Text data? If yes, then how. If no, then why?

Yes, CNN can be applied to text data.

In text processing, the text is first converted into numerical representations such as word embeddings (e.g., Word2Vec, GloVe) or token vectors. These vectors form a matrix where each row represents a word in the sentence.

How CNN works on text:

1. The text is converted into word embeddings.
2. The embeddings create a matrix representation of the sentence or document.
3. Convolution filters slide over the matrix to capture important patterns like phrases or n-grams.
4. Pooling layers select the most important features.
5. The extracted features are passed to fully connected layers for classification.

|                                                                                                                |                                                                                                                            |                                  |
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c. What is the role of dropout layer?

Neural networks employ the Dropout layer as a regularization strategy to lessen overfitting during training. During each training phase, a portion of the neurons in a layer are randomly disabled (dropped).

By doing this, the model is compelled to acquire more robust and generalized properties since it can no longer rely on particular neurons.

d. What will happen if maxpooling is replaced with minpooling?

The pooling layer will choose the least value from each feature map region rather than the maximum value if MaxPooling is swapped out for MinPooling.

Impact:

- Because MinPooling retains the smallest values, significant characteristics like strong edges and patterns may be lost.
- The model could concentrate on background or less important elements.
- This may result in less accurate models and worse feature extraction.